

Incident Handler's Journal

Maximilian Rivera | Security Portfolio | June 2026

JOURNAL ENTRIES

Date	June 2, 2026
Entry	1
Description	Reviewed a ransomware incident at a small healthcare clinic. Staff were locked out of systems for the full business day. Covers Detection and Analysis and Containment, Eradication, and Recovery phases.
Tool(s) used	No tools used. Entry based on incident report review.
The 5 W's	Who: An organized group of financially motivated hackers. What: Ransomware encrypted clinic files and demanded payment for the decryption key. When: Tuesday morning, approximately 9:00 AM. Where: A small healthcare clinic in the United States. Why: Healthcare was targeted because disrupted patient care creates pressure to pay fast.
Notes	No network segmentation and no offline backups meant the ransomware had access to everything once it got in. A basic backup strategy alone would have changed the recovery timeline significantly. This was not a sophisticated attack. It succeeded because of missing fundamentals, not attacker skill.

Date	June 5, 2026
Entry	2
Description	Investigated a spear phishing email targeting the CFO of Imaginary Bank. Analyzed the sender domain, body content, and download links to determine legitimacy. Covers Detection and Analysis.
Tool(s) used	Manual header analysis and URL inspection.
The 5 W's	Who: Unknown threat actor impersonating Imaginary Bank's board of directors. What: Phishing email pushing the CFO to download a fake tool called ExecuTalk, which would install malware. When: Saturday, December 21, 2019, at 3:05 PM. Where: CFO's corporate inbox at Imaginary Bank. Why: Attacker used a believable pretext and a 48-hour deadline to push the target into acting before thinking.
Notes	The domain alone flagged it — email came from @gmail.org, not @imaginarybank.com. Download links led to my.site.net/pwnexecs/, a credential harvesting page made to look like a legitimate login. Branding was convincing but the URL gave it away. Email was quarantined. Always verify the sender domain and hover over links before clicking.

Date	June 10, 2026
Entry	3
Description	Explored Suricata as a network-based IDS/IPS. Reviewed rule structure, tested alert output, and examined EVE JSON log format. Relevant to Detection and Analysis.
Tool(s) used	Suricata (network intrusion detection and prevention system)
The 5 W's	N/A — tool exploration entry.
Notes	Suricata rules have three parts: action (alert, pass, drop, reject), header (protocol, IPs, ports, direction), and rule options (content, flow, sid, rev). The flow keyword filters traffic by direction, which cuts noise when you only care about client-to-server requests. All output defaults to EVE JSON format, which plugs directly into a SIEM without parsing raw strings. Understanding rule structure is what lets an analyst reduce false positives instead of just living with them.

Date	June 12, 2026
Entry	4
Description	Used Wazuh to investigate failed SSH login attempts on Buttercup Games' mail server. Queried indexed log data to identify volume and pattern of root login failures. Covers Detection and Analysis.
Tool(s) used	Wazuh (SIEM platform)
The 5 W's	N/A — tool exploration entry.
Notes	Query: host.keyword: mailsrv AND (fail* OR failed) AND root returned 300+ events targeting the root account. The wildcard on fail* caught all variations of the word across different log formats. 300+ failed root SSH attempts in a single window is an automated brute force attack, not random noise. The data was there the whole time. Without a structured query, finding this pattern manually would have taken hours.

REFLECTIONS / NOTES

Were there any specific activities that were challenging for you?

Packet analysis was the hardest. Reading tcpdump output and mapping fields to the TCP/IP model required a completely new mental model. It clicked once I stopped trying to read the output like text and started treating each field as a structured piece of data with a specific location.

Has your understanding of incident detection and response changed?

Significantly. I came in thinking detection was mostly about having the right tools. What I actually learned is that tools are only as useful as the queries and rules behind them. The NIST Incident Response Lifecycle gave me a repeatable structure to apply instead of treating each incident as a new problem.

Was there a specific tool or concept you enjoyed the most?

SQL for security investigations. I already had a background working with structured data in supply chain and compliance, so applying that to log analysis felt natural. Writing queries to isolate specific login behavior or flag access patterns was one of the more immediately practical things I learned in this course.